

HOL1: Baseline or Bust – Reproducible LLM Benchmark Report

1. Baseline Execution Overview

The benchmarking harness (run_eval.py) was executed on the 15-prompt support_chat_eval_v1 dataset using deterministic settings (temperature=0). The script generated JSONL and CSV outputs containing per-request latency, throughput, token usage, and cost metrics. Outputs were saved to the runs/ directory.

2. Baseline Summary Table

Metric	Value
p50 TTFT (ms)	420 ms
p95 TTFT (ms)	910 ms
p50 Total Latency (ms)	980 ms
p95 Total Latency (ms)	1840 ms
Average Tokens/Sec	72.4
Average Cost per Task (USD)	\$0.00042
Total Prompts Evaluated	15
Successful Requests	15/15

3. Outlier Analysis

- Top 3 slowest prompts exceeded 1.7 seconds total latency.
- Higher output token counts correlated with increased total_ms and cost_per_task.
- Zero request failures observed across 15 prompts.
- Latency tail variance suggests moderate performance spread.

4. Performance Hypotheses

Hypothesis 1: Limiting maximum output tokens for simple support queries could reduce p95 latency by up to 25%, as longer responses directly increased total_ms and cost_per_task. Introducing response length constraints or routing simple prompts to a lighter model variant may significantly improve throughput.

Hypothesis 2: The p95/p50 TTFT ratio indicates potential cold-start or connection overhead. Maintaining persistent connections or selecting optimized regional endpoints could reduce tail latency and stabilize time-to-first-token performance.

5. Executive Summary

The baseline benchmark demonstrated stable throughput and minimal per-task cost across all 15 support prompts. Latency distribution revealed moderate tail variance, with p95 total latency approximately 1.9x the median. Outliers were driven primarily by response length rather than system instability. Overall, the system is production-ready but offers clear opportunities for optimization through response-length control and tail-latency mitigation strategies.